system impacts users' belief in established facts and resistance to factual evidence. This is particularly important as mitigation techniques such as labels and watermarks to establish the provenance of AI-generated content become more widespread. Evidence suggests that labelling content as synthetic or "fake" could have unwanted consequences, such as increasing the perceived accuracy and believability of unlabelled content (otherwise known as the "implied truth effect") (Pennycook et al., 2020), and that these effects vary based on partisanship and label origin (Jia et al., 2022). These findings should be rigorously tested in the context of generative AI systems.

A.3.3. Systemic impacts

Operationalisation: Public trust in media and information. Societal harms can be anticipated in foresight exercises and mitigated before they occur, but they can only be empirically measured at the third layer, once the AI system is deployed in society. This is because they require a certain amount of take-up and time before the effects manifest. To evaluate whether AI system adoption contributes to a broader erosion of trust at scale, population-level analysis of shifts in public trust can be conducted. Similar studies have been done in the context of social media and trust in mainstream information sources. Park et al. (2020), for example, explore the impact of social media use on trust in news media globally. Lorenz-Spreen et al. (2023) find strong correlational and causal evidence of a relationship between digital media use and low political trust, i.e. trust in media and political institutions.

Operationalisation: Prevalence of synthetic content. One way to test whether AI system adoption threatens to "pollute" and reduce the quality of publicly available information would be to measure the prevalence and spread of false or misleading synthetic content in the public domain post deployment. Allen et al. (2020), for example, evaluate the prevalence of fake news at the scale of the US information ecosystem, using nationally representative samples of mobile, desktop, and television-based media consumption – an approach that could be usefully extended to synthetic data. To test the impact of deploying a generative AI system on human-generated data on the open web, del Rio-Chanona et al. (2023) compared activity on Stack Overflow in locales with different levels of access to the AI system.

Operationalisation: Prevalence of synthetic content. Another way to evaluate the likelihood of synthetic content contaminating the public domain at this layer is to test the effectiveness of methods for identifying AI-generated content, such as watermarking. Text-based watermarking is accomplished by adding hidden signals, such as tagging a subset of words, to the generated text (Kirchenbauer et al., 2023). For images and video, watermarks work by adding an imperceptible perturbation to an image's pixel, creating an identifiable marker.[19]

AI systems used to create synthetic visual data embed unique traces in the output image, which can also be used to aid detection of fake images and attribute them to their different source AI systems (Sha et al., 2023). AI labs – including Open AI, Google DeepMind, and HuggingFace, and public agencies like DARPA (e.g. through its SemaFor programme) – are currently developing automatic tools to help developers, key stakeholders, and the public better detect synthetic media. However, existing generative AI detection tools often struggle with generalisation, perform poorly on languages other than English, and are ill-equipped to deal with synthetic media that have been compressed or resized, as is common on social media (Corvi et al., 2022; Liang et al., 2023). Watermarks are vulnerable to simple manipulation like resizing, cropping, or simply changing content format. These techniques therefore have to be continuously updated to prevent misuse.

[19]There have also been growing efforts around authentication of human-generated audiovisual content. See, for example, the work led by the Coalition for Content Provenance and Authenticity (C2PA).

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